

Part 3: Gradient Descent Manual Calculation

Objective: Manually compute updates of the gradient descent algorithm for the parameters m and b in a simple linear regression model.

Given:

- Linear equation: $y = mx + b$
- Initial $m = -1$
- Initial $b = 1$
- Learning rate $\alpha = 0.1$
- Given points: $(1, 3)$ and $(3, 6)$

Formulas:

- Predicted value: $\hat{y}_i = mx_i + b$
- Gradient of the cost function (Mean Squared Error - MSE):

$$\frac{\partial J}{\partial m} = -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i)x_i$$

$$\frac{\partial J}{\partial b} = -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

- Update rules:

$$m_{new} = m_{old} - \alpha \frac{\partial J}{\partial m}$$
$$b_{new} = b_{old} - \alpha \frac{\partial J}{\partial b}$$

Here, n is the number of data points, $n = 2$.

Iteration 1

Initial values: $m = -1$

$b = 1$

Data points:

Point 1: $(x_1, y_1) = (1, 3)$

Point 2: $(x_2, y_2) = (3, 6)$

Step 1: Compute predicted values $\hat{y}_i$

For Point 1 $(1, 3)$:

$$\hat{y}_1 = mx_1 + b = (-1)(1) + 1 = -1 + 1 = 0$$

For Point 2 $(3, 6)$:

$$\hat{y}_2 = mx_2 + b = (-1)(3) + 1 = -3 + 1 = -2$$

Step 2: Compute the gradients $\frac{\partial J}{\partial m}$ and $\frac{\partial J}{\partial b}$

Calculate $(y_i - \hat{y}_i)$ for each point:

For Point 1: $y_1 - \hat{y}_1 = 3 - 0 = 3$

For Point 2: $y_2 - \hat{y}_2 = 6 - (-2) = 6 + 2 = 8$

Calculate $\frac{\partial J}{\partial m}$:

$$\sum_{i=1}^n (y_i - \hat{y}_i)x_i = (3)(1) + (8)(3) = 3 + 24 = 27$$

$$\frac{\partial J}{\partial m} = -\frac{2}{2}(27) = -27$$

Calculate $\frac{\partial J}{\partial b}$:

$$\sum_{i=1}^n (y_i - \hat{y}_i) = 3 + 8 = 11$$

$$\frac{\partial J}{\partial b} = -\frac{2}{2}(11) = -11$$

Step 3: Update m and b

$$m_{new} = m_{old} - \alpha \frac{\partial J}{\partial m} = -1 - (0.1)(-27) = -1 + 2.7 = 1.7$$

$$b_{new} = b_{old} - \alpha \frac{\partial J}{\partial b} = 1 - (0.1)(-11) = 1 + 1.1 = 2.1$$

Results of Iteration 1:

$$m = 1.7$$

$$b = 2.1$$

Iteration 2

Current values:

$$m = 1.7$$

$$b = 2.1$$

Step 1: Compute predicted values $\hat{y}_i$

For Point 1 (1, 3):

$$\hat{y}_1 = mx_1 + b = (1.7)(1) + 2.1 = 1.7 + 2.1 = 3.8$$

For Point 2 (3, 6):

$$\hat{y}_2 = mx_2 + b = (1.7)(3) + 2.1 = 5.1 + 2.1 = 7.2$$

Step 2: Compute the gradients $\frac{\partial J}{\partial m}$ and $\frac{\partial J}{\partial b}$

Calculate $(y_i - \hat{y}_i)$ for each point:

For Point 1: $y_1 - \hat{y}_1 = 3 - 3.8 = -0.8$

For Point 2: $y_2 - \hat{y}_2 = 6 - 7.2 = -1.2$

Calculate $\frac{\partial J}{\partial m}$:

$$\sum_{i=1}^n (y_i - \hat{y}_i)x_i = (-0.8)(1) + (-1.2)(3) = -0.8 - 3.6 = -4.4$$

$$\frac{\partial J}{\partial m} = -\frac{2}{2}(-4.4) = 4.4$$

Calculate $\frac{\partial J}{\partial b}$:

$$\sum_{i=1}^n (y_i - \hat{y}_i) = -0.8 - 1.2 = -2.0$$

$$\frac{\partial J}{\partial b} = -\frac{2}{2}(-2.0) = 2.0$$

Step 3: Update m and b

$$m_{new} = m_{old} - \alpha \frac{\partial J}{\partial m} = 1.7 - (0.1)(4.4) = 1.7 - 0.44 = 1.26$$

$$b_{new} = b_{old} - \alpha \frac{\partial J}{\partial b} = 2.1 - (0.1)(2.0) = 2.1 - 0.2 = 1.9$$

Results of Iteration 2:

$$m = 1.26$$

$$b = 1.9$$

Iteration 3

Current values:

$$m = 1.26$$

$$b = 1.9$$

Step 1: Compute predicted values $\hat{y}_i$

For Point 1 (1, 3):

$$\hat{y}_1 = mx_1 + b = (1.26)(1) + 1.9 = 1.26 + 1.9 = 3.16$$

For Point 2 (3, 6):

$$\hat{y}_2 = mx_2 + b = (1.26)(3) + 1.9 = 3.78 + 1.9 = 5.68$$

Step 2: Compute the gradients $\frac{\partial J}{\partial m}$ and $\frac{\partial J}{\partial b}$

Calculate $(y_i - \hat{y}_i)$ for each point:

For Point 1: $y_1 - \hat{y}_1 = 3 - 3.16 = -0.16$

For Point 2: $y_2 - \hat{y}_2 = 6 - 5.68 = 0.32$

Calculate $\frac{\partial J}{\partial m}$:

$$\sum_{i=1}^n (y_i - \hat{y}_i) x_i = (-0.16)(1) + (0.32)(3) = -0.16 + 0.96 = 0.80$$

$$\frac{\partial J}{\partial m} = -\frac{2}{2}(0.80) = -0.80$$

Calculate $\frac{\partial J}{\partial b}$:

$$\sum_{i=1}^n (y_i - \hat{y}_i) = -0.16 + 0.32 = 0.16$$

$$\frac{\partial J}{\partial b} = -\frac{2}{2}(0.16) = -0.16$$

Step 3: Update m and b

$$m_{new} = m_{old} - \alpha \frac{\partial J}{\partial m} = 1.26 - (0.1)(-0.80) = 1.26 + 0.08 = 1.34$$

$$b_{new} = b_{old} - \alpha \frac{\partial J}{\partial b} = 1.9 - (0.1)(-0.16) = 1.9 + 0.016 = 1.916$$

Results of Iteration 3:

$$m = 1.34$$

$$b = 1.916$$

Iteration 4

Current values:

$$m = 1.34$$

$$b = 1.916$$

Step 1: Compute predicted values $\hat{y}_i$

For Point 1 (1, 3):

$$\hat{y}_1 = mx_1 + b = (1.34)(1) + 1.916 = 1.34 + 1.916 = 3.256$$

For Point 2 (3, 6):

$$\hat{y}_2 = mx_2 + b = (1.34)(3) + 1.916 = 4.02 + 1.916 = 5.936$$

Step 2: Compute the gradients $\frac{\partial J}{\partial m}$ and $\frac{\partial J}{\partial b}$

Calculate $(y_i - \hat{y}_i)$ for each point:

For Point 1: $y_1 - \hat{y}_1 = 3 - 3.256 = -0.256$

For Point 2: $y_2 - \hat{y}_2 = 6 - 5.936 = 0.064$

Calculate $\frac{\partial J}{\partial m}$:

$$\sum_{i=1}^n (y_i - \hat{y}_i) x_i = (-0.256)(1) + (0.064)(3) = -0.256 + 0.192 = -0.064$$

$$\frac{\partial J}{\partial m} = -\frac{2}{2}(-0.064) = 0.064$$

Calculate $\frac{\partial J}{\partial b}$:

$$\sum_{i=1}^n (y_i - \hat{y}_i) = -0.256 + 0.064 = -0.192$$

$$\frac{\partial J}{\partial b} = -\frac{2}{2}(-0.192) = 0.192$$

Step 3: Update m and b

$$m_{new} = m_{old} - \alpha \frac{\partial J}{\partial m} = 1.34 - (0.1)(0.064) = 1.34 - 0.0064 = 1.3336$$

$$b_{new} = b_{old} - \alpha \frac{\partial J}{\partial b} = 1.916 - (0.1)(0.192) = 1.916 - 0.0192 = 1.8968$$

Results of Iteration 4:

$m = 1.3336$

$b = 1.8968$

Summary of Results and Trend Observation

Iteration $\frac{\partial J}{\partial b}$	m	b	$\hat{y}_1$ (for (1,3))	$\hat{y}_2$ (for (3,6))	$y_1 - \hat{y}_1$	$y_2 - \hat{y}_2$	$\frac{\partial J}{\partial m}$
0 (Initial) -11	-1	1	0	-2	3	8	-27
1 2.0	1.7	2.1	3.8	7.2	-0.8	-1.2	4.4
2 -0.16	1.26	1.9	3.16	5.68	-0.16	0.32	-0.80
3 0.192	1.34	1.916	3.256	5.936	-0.256	0.064	0.064
4 0.192	1.3336	1.8968	3.256	5.936	-0.256	0.064	0.064

The trend continues to show the magnitudes of the gradients decreasing, which strongly indicates that the model is converging towards a minimum error. The values of m and b are making smaller adjustments with each iteration, suggesting they are approaching their optimal values for the given data.